Fiscal Year:	FY 2023	**Task Last Updated:**	FY 11/08/2022
PI Name:	Wood, Scott J. Ph.D.		
Project Title:	Recovery of Functional Performance Following Long Duration Space Flight (Field Test)		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:	NOTE: PI returned to NASA JSC in January 2017. PI was at Azusa Pacific University from August 2013 – January 2017; prior to August 2013, PI was at NASA JSC.		
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	Directed Research
Start Date:	08/26/2022	**End Date:**	09/30/2022
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
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No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
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Flight Program:	PostFlight		
Flight Assignment:	ISS Postflight studies. NOTE: Project start date updated due to P.I. change to S. Wood. Information per C. Ribeiro/NASA-JSC. (Ed. 8/29/22).		
Key Personnel Changes/Previous PI:	NOTE: Marissa Rosenberg, Ph.D. has left NASA and is no longer with the project (Ed. 8/29/22).		
COI Name (Institution):	Tomilovskaya, Elena Ph.D. (Institute of Biomedical Problems, Moscow) Lee, Stuart Ph.D. (KBR, NASA Johnson Space Center Cardiovascular and Vision Laboratory)		
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Task Description:	The Field Test (FT) proposal represents a joint effort between the Neuroscience and Cardiovascular Laboratories at the Johnson Space Center and the Institute of Biomedical Problems Sensorimotor Laboratory and Cardiovascular Laboratory, Moscow, Russia. The primary goal of this proposal is to determine functional performance in long-duration space flight crews beginning as soon after landing as possible (<2 hr) with one to three immediate follow-up measurements on the day of landing. This goal has both sensorimotor and cardiovascular elements with an evaluation of NASA's new compression garment with the Russian traditional Kentavr garment. In addition to the immediate post-landing collection of data, post-flight data was acquired at several timepoints on landing day to characterize recovery to preflight baseline levels. Clearly measurable performance parameters such as the ability to perform a seat egress, recover from a fall or the ability to see clearly when walking, and related physiological data (orthostatic responses) are required to provide an evidence base for characterizing programmatic risks and variability among crewmembers. Overall, these early functional and related physiological measurements will allow for the establishment of a sensorimotor and cardiovascular recovery time constant that has not been previously captured in over 50 years of space flight. Specific Aims: 1. Quantify functional performance from measurements on long-duration crewmembers taken as close in time to landing as possible. 2. Develop a recovery timeline of functional performance on long-duration crewmembers. 3. Determine the efficacy of U.S. and Russian compression garments as countermeasures for alleviating orthostatic intolerance.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Vestibular and sensorimotor alterations represent one of the greatest clinical challenges impacting crew activities following G-transitions. In order to capture the initial decrements in performance, we successfully developed a portable set of measures and recording instrumentation that was compatible with relatively harsh environments. The fall risks on Earth associated with vestibular and sensorimotor impairment are underestimated largely due to the lack of testing available to the general population. The National Institutes of Health has been actively funding efforts to develop a Toolbox of field measures for the vestibular, vision, and motor sensory domains. We propose that the portable measures validated during our Field Tests will be beneficial for the broader characterization of how vestibular and sensorimotor deficits contribute to fall risks on Earth. The results of our Aim 3 to evaluate compression garment efficacy could have potential application to patients with a history of orthostatic intolerance. For example, abdominal and lower body compression reduced heart rate and improved symptoms during had upright tilt in adult patients with postural orthostatic tachycardia syndrome (POTS, Bourne et al, 2021).
	Since the data collection was completed in April 2021, during this last reporting period we continued data analysis, published an additional manuscript (Rosenberg et al., Front Physiol, 2022), and are in the process of completing the final manuscripts. A brief summary of some key measures is presented below. The NASA and Russian teams collected data on a total of 48 United States Orbital Segment (USOS) and Russian Space Agency (RSA) crewmembers tested over 26 Soyuz landings. Eighteen crewmembers (7 RSA, 11 USOS) completed the reduced Pilot Field Test (FT) protocol and 30 crewmembers (29 RSA and 11 USOS) participated in the full FT. Nine of the RSA full FT subjects had previously participated in either the pilot FT (n=5) or full FT (n=4), so 39 different crewmembers were included in the total sample. Aims 1 and 2: Recovery of functional performance Since the tasks were designed to capture generic functional elements of critical mission tasks, the inability to perform the task, generally due to motion sickness and/or extreme motion sensitivity, was considered an important metric of a deconditioned long duration crewmember population. Eight out of 48 subjects (16.7%) were unable to attempt any testing at the initial R+0a opportunity either in the medical tent or nearby rally airport. An additional 17 out of 48 subjects (35.4%) were unable to complete all trials on R+0a. Motion Sickness: On landing day, 100% of crewmembers reported some level of landing-related motion sickness symptoms (1-20 scale with 1=no symptoms and 20=vomiting). Based on motion sickness reports from 45 crewmembers obtained on R+0a, the maximum motion sickness score during testing on R+0a was on average 13.2 (± 5.1 std). For the 19 USOS crew that reported motion sickness across all three landing sessions, their motion sickness score decreased with each subsequent session: R+0a was 12.9 ± 5.8, R+0b was 8.9 ± 5.8, and R+0c was 5.4 ± 4.3 (mean ± std). Sit-to-Stand: Standing up from a seated position is one of the most frequently performed functional tasks and is an essential prerequisite to walking and capsule egress. Based on the single exponential model, the estimated time of 4.0 s required to complete the task immediately following landing was more than double the preflight median score of 1.8 s. However, there was considerable variability with times to stability ranging between 1.25-6.28 s within the initial four hours. The recovery was relatively quick on this simple task with a time constant of 21.4 h. Recovery from Fall/Prone-to-Stand: The prone-to-stand maneuver has been used by our laboratory to combine postural stability and orthostatic challenges to examine vestibular-autonomic interactions following spaceflight. The cardiovascular findings during this test are summarized below under compression garment efficacy. Times to complete this task increased from a preflight median score of 7.9 s to 17.4 s at time = 0 according to the single exponential fit. Responses ranged between 7.6 – 27.2 s within the initial four hours. The recovery was similar to but slightly longer than the sit-to-stand test, with a time constant of 21.9 h. Tandem Walk: Tandem walking has been an integral part of dynamic gait testing for neurological exams both for returning astronauts and for a variety of disorders including cerebellar disease, vestibular pathology, and neurodegenerative conditions. Preflight, the eyes open median score for correct steps was 100% where the eyes closed median score was 88.3% with increased intersubject variability. The initial scores in the medical tent did not vary between these two tasks. Based on the single exponential fits, the percentage of correct steps between these two conditions was 11.7% with eyes open and 10.1% with eyes closed. Responses within the initial four hours ranged between 0-78% with eyes open and between 0-56% with eyes closed. One can infer from these comparisons that vision was not sufficient to compensate for the early postflight vestibular disruption on this type of challenging dynamic gait task. While tandem walking with eyes open had a time constant of 21.3 h, the time constant increased to 86.8 h with eyes closed.

Task Progress:	Obstacle Walk: The clinical Timed "Up & Go" test was modified during Field Test by the introduction of small obstacles (5-15 cm in height) in the walking path to provide a field version of the seated egress task that had been used in previous studies. As previously published by Reschke et al. (Aerosp Environ Med, 2020), the time to completion was significantly increased on landing day for all three obstacle heights. The differences were greatest for the first trial using the 5 cm obstacle ($t/37 = 10.5$, $p < 0.01$) but were also highly significant for the 10 cm ($t/37 = 8.0$, $p < 0.01$) and 15 cm obstacles ($t/37 = 7.6$, $p < 0.01$). The greater difference for the 5 cm obstacle was likely because this was the first trial. The use of inertial sensors allowed evaluation of turning parameters around the cone. The baseline preflight turn rate was 134.36 ± 21.17°/s, with rates ranging from 58.14 to 191.20 °/s. However, on R+0a turn rates decreased to an average of 68.19 ± 15.10°/s, with a maximum turn rate of 95.39°/s, highlighting the difficulty of turning at that early timepoint. Potential Effects of Multiple Field Tests on Landing Day: Although the protocol was challenging and temporarily increased motion sickness symptoms, there were anecdotal reports that performing these tasks within the first few hours of landing accelerated their recovery. Therefore, results from computerized dynamic posturography (CDP) following return to Houston were used to compare recovery between crewmembers that participated in FT (n=18) with those that did not (controls, n=11). As reported in Rosenberg et al. (Front Physiol, 2022), it was evident that some FT participants tended to perform closer to, or even better, than their preflight baselines. Nevertheless, the distribution of difference scores in the FT group, particularly for SOT-5M, appeared bimodal and suggests that FT participation did not improve recovery in all subjects, illustrating the importance of maintaining activity within an individual's threshold. AIM 3: Gradient Compression Garment Efficacy The primary focus of Aim 3 was to determine the efficacy of the gradient compression garment (GCG) to prevent increases in heart rate and prevent the development of signs of presyncope on landing day after a long-duration spaceflight. Results from the assessment of the GCG during the stand test on R+0 have been published previously (Lee et al., Front Physiol, 2020), but are briefly summarized here. All astronauts completed the stand test preflight, but three astronauts were unable to attempt the stand test at R+0a and one of these was unable to start the test at R+0b. Of the astronauts who started the stand test, only one astronaut was unable to complete 3.5 min of standing at R+0b (stand test terminated at 3.3 min). Of the astronauts who participated in testing on landing day, the heart rate and mean arterial pressure responses to standing (stand-prone) were not different than preflight at any of the postflight sessions (data not shown). The heart rate response to standing at R+0a was significantly less than that observed in test subjects after 70 days of bedrest and at R+0c was less than that observed in previous reports in astronauts one day after landing. In conclusion, the GCG on R+0 prevented the tachycardia that normally occurs while standing after spaceflight without compression garments and protected against a decrease in blood pressure. This past year we combined all the data from the Pilot Field Test and Field Test Studies to use a larger number of subjects to assess the efficacy of the lower body compression garments to protect against the signs of orthostatic intolerance on landing day. In this case, we compared preflight data for all the subjects to the data collected on R+0a, whether the crewmembers were wearing the GCG or the Kentavr. As in our previous analyses, the heart rate response to standing was not different than preflight in the GCG subjects (+2 bpm, $p=0.88$), but the heart rate response to standing in the Kentavr subjects was greater than that measured during preflight testing (+10 bpm, $p<0.001$). Further, mean arterial pressure was less well protected at R+0a in crewmembers wearing the Kentavr; the response of mean arterial pressure to standing was significantly less on R+0a when wearing the Kentavr (-10 mmHg, $p<0.001$) than preflight without the garment. We have previously demonstrated that the Kentavr is effective at preventing an increase in heart rate during upright posture in a ground-based study of hypovolemic subjects, and thus we suspect that the elevated heart rate recorded in crewmembers wearing the Kentavr postflight reflects a situation in which the device was not adjusted to maximize its efficacy. The final objective of Aim 3 was to characterize the recovery of stand test responses in the weeks after landing. From our previous experience with long-duration spaceflight, it is clear that crewmembers rapidly adapt to the 1-G environment; the prevalence of orthostatic intolerance is much less on R+1, and plasma volume is restored through the combination of endocrine responses, intravenous fluid administration, and oral fluid intake. Surprisingly, the heart rate response to standing on R+3/4 without the Kentavr was not different than preflight (+2 bpm, $p=0.81$) but was greater than preflight on R+6/8 (+ 6 bpm, $p=0.01$) and on R+12 (+6 bpm, $p=0.01$). However, the mean arterial pressure response to standing was not different than preflight on R+3 (+1 mmHg, $p=0.99$), R+7 (+2 mmHg, $p=0.99$), and R+10/12 (+2 mmHg, $p=0.49$). These results may be influenced by the crewmembers who were not included in the analyses because they were wearing the Kentavr on the day of testing. Some astronauts and cosmonauts continue to wear all or parts of the Kentavr for several days after landing to manage orthostatic intolerance symptoms as well as lower leg swelling. For the purposes of this report, we did not categorize subjects as "wearing the Kentavr" unless they were wearing, at the least, the shorts. Previous experience in our laboratory (unpublished data) suggests that wearing the calf-only compression garments, like the Kentavr gaiters, has limited effects on orthostatic responses. Presumably crewmembers who continued to wear the Kentavr on R+3 (n=5), R+7 (n=2), and R+12 (n=2) and were not considered in these analyses would have had greater heart rate responses to standing had they not been wearing the garment and perhaps would have experienced symptoms of orthostatic intolerance.
Bibliography Type:	Description: (Last Updated: 02/20/2026)
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